

Suppose $HotelCosts = [0, 1, 4, 3, 2, 0]$. At which mile marker(s) does an optimal trip stop?

2, 5 ~~1, 2~~ 4 3

What is the cost of the optimal trip from (a)? 3

Complete the English recursive subproblem definition for this problem.

Let $MinCost(i)$ denote... the min cost for momo to travel from mile marker 1 to mile marker i , staying at hotel i .

For what value of i does $MinCost(i)$ give the solution to the full problem? n

Fill in the recurrence for this problem. You may not need every row given.

$MinCost(i) = \begin{cases} \text{HotelCost}[2] & \text{if } i=1 \\ \text{HotelCost}[3] & \text{if } i=2 \\ \text{HotelCost}[i] + \min(MinCost(i-1), MinCost(i-2), MinCost(i-3)) & \text{if } i=3 \\ \text{HotelCost}[i] + \min(MinCost(i-1), MinCost(i-2), MinCost(i-3)) & \text{if } i > 3 \end{cases}$

6. Complete the pseudocode for a memoized algorithm to solve the problem.

MinCost 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

ComputeMinTotalCost($HotelCosts[1..n]$)

let array $MinCost[1..n]$

$MinCost[1] = 0$

$MinCost[2] = HotelCosts[2]$

$MinCost[3] = HotelCosts[3]$

for i in $4, \dots, n$:

$MinCost[i] = HotelCosts[i] +$

$\min(MinCost[i-1], MinCost[i-2], MinCost[i-3])$

return $MinCost[n]$